

TASK 2: SENSOR-BASED IoT SIMULATION

CodeAlpha – Internet of Things (IoT) Internship

Project: Automatic LED Control Using a Light Sensor

Objective

The objective is to simulate a simple IoT-style sensor system in which a light sensor provides an input and an LED acts as an output. The system demonstrates the basic idea of sensing an environmental condition and taking an automated action. CodeAlpha specifically allows a sensor-based simulation using Tinkercad or Proteus, with an example such as LED on/off using a temperature, light or motion sensor.

Components

- Arduino Uno
- LDR (light-dependent resistor)
- 10 k Ω resistor
- LED
- 220 Ω resistor
- Breadboard and jumper wires

Working Principle

The LDR and resistor form a voltage divider connected to an Arduino analog input. The Arduino reads the sensor value. If the light level crosses the selected threshold, the program changes the LED state. This is a simple sensor-to-controller-to-output IoT concept.

Arduino Code

```
int sensorPin = A0;
int ledPin = 8;
int threshold = 500;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);

  if (sensorValue < threshold) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
  }
  delay(200);
}
```

Simulation Connections

1. Connect the LDR voltage-divider output to Arduino A0.
2. Connect the fixed resistor to complete the voltage-divider circuit.
3. Connect Arduino digital pin 8 to the LED through a 220 Ω resistor.
4. Connect the LED return to GND.
5. Connect all required grounds together.

Expected Output

When the simulated light condition changes, the analog sensor reading changes. When the reading goes below the selected threshold, the LED turns ON; otherwise, it turns OFF. The Serial Monitor can be used to observe the sensor values.

Simulation Screenshot

Important: The CodeAlpha instructions require simulation screenshots plus a code explanation. I have not inserted a fake Tinkercad/Proteus screenshot. After running this circuit in Tinkercad or Proteus, add one screenshot showing the circuit and one showing the output/Serial Monitor to this PDF before submission.

Code Explanation

The program reads the LDR value through analog pin A0. The threshold variable defines the switching point. The if-else condition controls the LED connected to digital pin 8. Serial output helps monitor the sensor reading during simulation.

Conclusion

This simulation demonstrates a basic sensor-based IoT system: a physical condition is sensed, the controller processes the reading, and an output device responds automatically. The same basic structure can be extended to larger IoT applications.

Reference

CodeAlpha, "Internet of Things (IoT) – Internship Overview and Task List," provided internship instructions.